NEELAY SHAH

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Website ◇ GitHub ◇ LinkedIn ◇ Google Scholar

EDUCATION

Birla Institute of Technology and Science Pilani, Goa, India 2018 - Present
Bachelor of Engineering (B.E. Hons.), Electronics and Instrumentation

EXPERIENCE

Ethical Intelligence Lab, Harvard Business School Jan 2021 - Present
Research affiliate (remote) | Mentor: Dr. Julian De Freitas

- Working on developing human-like and human-aware autonomous driving algorithms
- Previously worked on developing deep neural networks to model human visual cognition of social interactions
- Tech used: Python, PyTorch, Pandas, scikit-learn

Centre of Intelligent Robotics, IIIT Allahabad June 2020 - July 2020
Research intern (remote) | Mentor: Dr. GC Nandi

- Used model compression techniques to develop compact multi-modal emotion recognition models for human-robot interaction using deep learning
- Implemented generative grasping convolutional neural networks (GGCNN) for robotic grasping
- Tech used: Python, PyTorch, OpenCV

Aurum Smart Tech May 2020 - June 2020
Software development intern (remote)

- Worked on developing an image processing pipeline for QR code scanning
- Tech used: Python, ZBar

PUBLICATIONS/PREPRINTS

KD-Lib: A PyTorch library for Knowledge Distillation, Pruning and Quantization
Het Shah, Avishree Khare, ***Neelay Shah****, Khizir Siddiqui*
arXiv preprint | [Paper] [Code]

OPEN-SOURCE SOFTWARE

KD-Lib
[Code] [Documentation]
Co-creator

- An open-source PyTorch library for making model compression research in deep learning more accessible
- Contains implementations of algorithms from 3 families of model compression research - Knowledge Distillation, Pruning, and Quantization
- Simple and intuitive API to facilitate the use of compression techniques for custom deep learning models with few lines of code

TECHNICAL SKILLS

Programming Languages	Python, C++, Java
Frameworks/Libraries	PyTorch, Keras, PyGeNN, scikit-learn, JAX, NumPy, Pandas, Matplotlib, OpenCV, Scipy
Operating Systems	Linux, Windows
Tools	Git/GitHub, GitHub Actions, Travis CI, AWS EC2, LaTeX, Vim

RESEARCH PROJECTS

Machine Learning for Satellite Navigation Feb 2020 - April 2020
Mentor: Mr. Abhijit Dey
Digital Communication Lab, BITS Pilani Goa

- Developed a deep learning pipeline to forecast irregularities in the ionosphere which cause fluctuations in satellite navigation signals
- Benchmarked machine learning models for classifying multi-path interference in satellite signals
- Tech used: Python, Keras, Pandas

RELEVANT COURSEWORK

University	Calculus, Linear Algebra, Probability and Statistics, Discrete Mathematics, Digital Design, Introduction to Cognitive Neuroscience, Signals and Systems, Control Systems, Data Structures and Algorithms, Object-oriented Programming
Online	Convolutional Neural Networks for Visual Recognition (Stanford CS231n), Machine Learning (Stanford CS229), Introduction to Reinforcement Learning (UCL)

PERSONAL PROJECTS

Transfer Learning Aided One-shot Classification | ***[Code]***

- Transfer learning to improve the performance of a Siamese neural network for one-shot recognition on the MNIST dataset

Spiking Neural Networks for Speech Recognition | ***[Code]***

- Biologically plausible algorithms of speech recognition using optimized neuronal network simulation software

Unsupervised Anomaly Detection | ***[Code]***

- Autoencoders to identify anomalies in chest X-rays

Deep Learning Optimizers | ***[Code]***

- Toy implementations of popular deep learning optimization techniques from scratch

Machine Learning for Network Forensics | ***[Code]***

- Machine learning based network forensic mechanism for botnet activities in the Internet of Things (IoT)

GANs for Robotic Grasp Detection | ***[Code]***

- Exploratory use of image-to-image translation generative adversarial networks (GANs) to generate grasp orientations for robotic grasping

Signboard Detection | ***[Code]***

- Detection of signboards containing text in vernacular languages using deep learning

MENTORSHIP

Instructor - Introduction to Deep Learning
Student-run course | Centre for Technical Education, BITS Pilani Goa

- Taught and created reference material for an introductory deep learning course intended for college freshmen

OTHER ACTIVITIES

Member
Society for Artificial Intelligence and Deep Learning

· Involved in reading and discussion sessions on artificial intelligence research

Member
University cricket team

· Participated in university level cricket tournaments

MISCELLANEOUS

Languages spoken	English, Hindi, Gujarati, Marathi, German (A1 level)
Interests	Sports (esp. Cricket and Football), International Relations, Journalism